

Assignment 2

Total Marks=20

1. Write a Java program that takes an integer array from the user and then allows the user to perform the following operations using a menu with the following options [Marks=2+2+1=5]

A. Find the sum of all elements B. Find the maximum element C. Search for a given element

2. Write a Java program to read an array of integers and, for each element, find the nearest greater element on its right. If no such element exists, print -1. Use only arrays. The time complexity for the program should not be more than $O(n)$. [Marks=1+4=5]

Hint: Use an additional array as a stack. Input: Array = {4, 5, 2, 25, 7, 8} Output: 5 25 25 -1 8 -1

3. Write a Java program to take two 2D arrays (matrices) from the user and perform the following operations using a menu: [Marks = 5]

- A. Addition of two matrices
- B. Subtraction of two matrices
- C. Multiplication of two matrices
- D. Division of two matrices
- E. Display both input matrices

4. Write a Java program to take a square 2D array (matrix) from the user and perform the following operations using a menu: [Marks = 5]

- A. Display the original matrix
- B. Find and display the transpose of the matrix
- C. Find and display the determinant of a 2×2 matrix
- D. Find and display the determinant of a 3×3 matrix